



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

PART TEST

28-01-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I & I(N)] : (SCORE-I)
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	A	B	C	A	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	C,D	A,C	B,C,D	A,C,D	A,B,C	B
SECTION-II	Q.	1	2	3	4	5	6
	A.	9.64	11.00	9.00	22.50	1.75	19.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	D	B	B	C	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C,D	C	A,D	A	A,C,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	12.10 to 12.40	250.00	6.00	5.00	3.00	420.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	A	C	C	B	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	A,B	A,B	A,C,D	B,C	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	1.00	4.00	1.00	1.50	8.00	43.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)
1. Ans (C)

Between faces LMNO and PQRS

$$dR = \frac{\rho dr}{t \frac{\pi}{2} r}; R = \frac{2\rho}{t\pi} \ln \frac{b}{a}$$

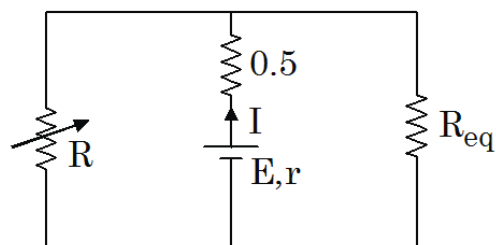
 Between faces LMRS and ONQP for an element at distance r from O.

$$dR = \frac{\rho \frac{\pi}{2} r}{t dr}$$

$$\text{So, } \frac{1}{R} = \int \frac{1}{dR} = \int_a^b \frac{t dr}{\rho \frac{\pi}{2} r} = \frac{2t}{\rho\pi} \ln \left(\frac{b}{a} \right)$$

$$R = \frac{\rho\pi}{2t \ln \left(\frac{b}{a} \right)}$$

2. Ans (A)

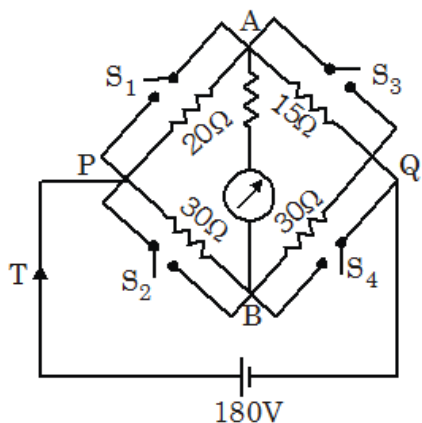
 If current through battery is maximum, then rate of chemical energy consumed through battery will be maximum. If $R = 0$ then current through battery will be maximum


$$\text{and } I_{\max} = \frac{E}{r + 0.5} = \frac{10}{0.5 + 0.5} = 10$$

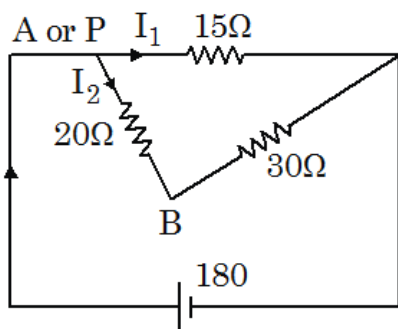
$$P_0 = EI = 10 \times 10 \text{ watt} = 100 \text{ watt}$$

9. Ans (B,C,D)

If S_1 closed. PB and AB in parallel



So effective circuit as shown



$$I_1 = \frac{180}{15} = 12 \text{ amp}; I_2 = \frac{180}{50} = 3.6 \text{ amp}$$

So $I = 15.6 \text{ amp}$.

When S_2 closed B is at higher potential than A so current in galvanometer from B to A.

When S_3 closed B is at higher potential than A so current in galvanometer from B to A.

When S_2 closed A is at higher potential than B so current in galvanometer from A to B

10. Ans (A,C,D)

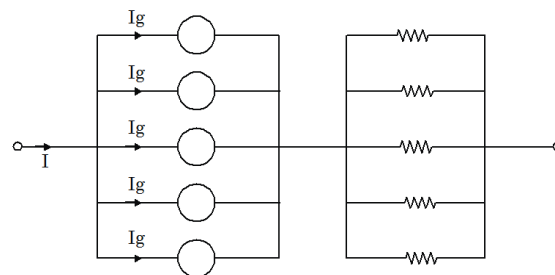
Situation (1) :



$$\text{Range} = I_g 10 R$$

$I_g \rightarrow$ maximum deflection of current galvanometer.

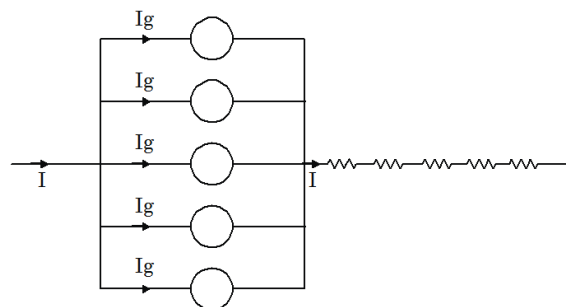
Situation (2) :



$$I = 5 I_g$$

$$\text{So Range} = 2 I_g R$$

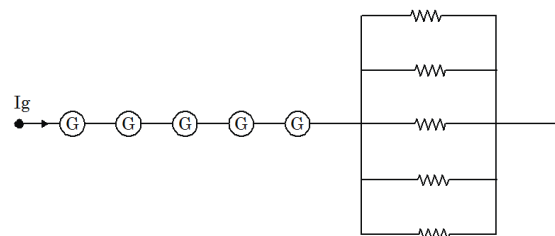
Situation (3) :



$$I = 5 I_g$$

$$\text{Range} = I_g R + 5 I_g 5 R = 26 I_g R$$

Situation (4) :



$$\text{Range} = 5 I_g R + I_g \frac{R}{5} = \frac{26 I_g R}{5} = 5.2 I_g R$$

11. **Ans (A,B,C)**

At any time t circuit is shown

$$I_1 = \frac{dQ_1}{dt} \dots\dots(1)$$

$$I_2 = \frac{dQ_2}{dt} \dots\dots(2)$$

$$\frac{Q_1}{C} + \frac{C\varepsilon - Q_2}{C} = \varepsilon$$

$$\frac{Q_1}{C} = \frac{Q_2}{C} \dots\dots(3)$$

So $I_1 = I_2$ from eqn. (3)

$$\frac{Q_1}{C} + 2I_1R = \varepsilon$$

$$I_1 = \frac{\varepsilon - \frac{Q_1}{C}}{2R} = \frac{C\varepsilon - Q_1}{2CR}$$

$$\frac{dQ_1}{dt} = \frac{C\varepsilon - Q_1}{2CR}$$

$$\int_0^{Q_1} \frac{dQ_1}{C\varepsilon - Q_1} = \int_0^t \frac{dt}{2CR}$$

$$-\ln(C\varepsilon - Q_1) \Big|_0^{Q_1} = \frac{t}{2RC}$$

$$Q_1 = C\varepsilon \left(1 - e^{-t/2RC}\right)$$

$$I_1 = \frac{dQ_1}{dt} = \frac{\varepsilon}{2R} e^{-t/2RC}$$

Current in $R = 2I_1$

Work done by battery

$$= Q_{\text{flow}} \varepsilon = C\varepsilon \cdot \varepsilon = C\varepsilon^2$$

12. **Ans (B)**

$$v_{\text{escape}} = \sqrt{\frac{2GM}{R}}$$

$$\% \text{ error in } v_{\text{escape}} = \frac{1}{2} \% \text{ error in } G$$

$$+ \frac{1}{2} \% \text{ error in } M$$

$$+ \frac{1}{2} \% \text{ error in } R$$

$$= \frac{1}{2} \times \left[\frac{0.0010}{6.6742} \times 100 + 0.01 + 0.02 \right]$$

$$= \frac{1}{2} [0.015 + 0.01 + 0.02]$$

$$= 0.0225$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (9.64)**

$$\text{Initial reading} = 3 + \frac{1}{50} \times 44 = 3.88 \text{ mm}$$

$$\text{Final reading} = 13 + \frac{1}{50} \times 26 = 13.52 \text{ mm}$$

So thickness of plate is

$$(13.52 - 3.88) \text{ mm} = 9.64 \text{ mm}$$

2. **Ans (11.00)**

Initial acceleration for large ball given by ma

$$= mg - F_B$$

$$V_1 d_1 \frac{g}{2} = V_1 d_1 g - V_1 \rho g$$

$[\rho = \text{density of water, } d_1 = \text{density of ball}]$

$$\Rightarrow d_1 = 2\rho$$

$$\text{Similarly, for second ball } d_2 = \frac{4\rho}{3}$$

After long time both of them will acquire terminal speed (u_{01} for larger ball and u_{02} for smaller ball)

$$6\pi\eta r u_{01} = mg - F_B$$

$$\Rightarrow 6\pi\eta r u_{01} = \frac{4}{3}\pi r^3 d_1 g - \frac{4}{3}\pi r^3 \rho g$$

$$\Rightarrow u_{01} = \frac{2}{9} \frac{r^2 \rho g}{\eta} \quad \text{Similarly, } u_{02} = \frac{r^2 \rho g}{54\eta}$$

$\therefore$ Velocity of smaller ball with respect to the larger ball is

$$= \left(\frac{1}{54} - \frac{2}{9} \right) r^2 \frac{\rho g}{\eta} = -\frac{11}{54} \frac{r^2 \rho g}{\eta} = \frac{11}{54} \frac{r^2 \rho g}{\eta} (\uparrow)$$

3. **Ans (9.00)**

Speed at open end of D

$$v_D = \sqrt{2g(0.03)}$$

$$v_{\text{at C}} = 2\sqrt{2g(0.03)}$$

$$\rho g h_2 = \frac{1}{2} \rho v_C^2 - \frac{1}{2} \rho v_D^2$$

$$\rho g h_2 = \frac{1}{2} \rho [3.2g(0.03)]$$

$$h_2 = 0.09 \text{ m or } 9 \text{ cm}$$

4. **Ans (22.50)**

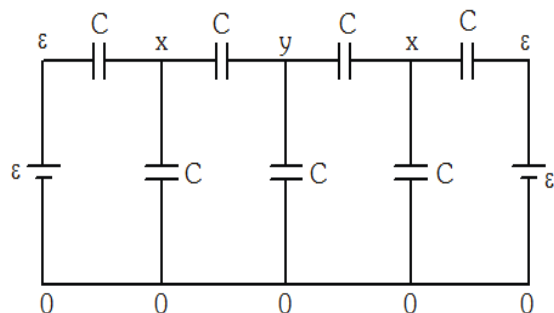
$$\frac{\varepsilon}{R} = 5$$

$$\frac{\varepsilon}{R+3} = 3$$

On solving reading of voltmeter is

$$\Delta V = 22.5 \text{ volt}$$

5. **Ans (1.75)**



Potential at different point shown in figure

By junction rule at

$$C(\varepsilon - X) = C(x - y) + C(x - 0)$$

$$\varepsilon = 3x - y \dots(1)$$

At (y)

$$2C(x - y) = (y - 0) C$$

$$2x = 3y \dots(2)$$

$$x = \frac{3\varepsilon}{7}, \quad y = \frac{2\varepsilon}{7}$$

$$\text{Charge through battery} = C(\varepsilon - x) = \frac{4C\varepsilon}{7}$$

$$\text{Work done by both the battery} = \frac{4C\varepsilon^2}{7} \times 2$$

$$\text{Heat} = \frac{1}{2} \text{ of work done by both battery}$$

$$= \frac{4C\varepsilon^2}{7}$$

6. **Ans (19.00)**

After closing S_1 charge on $C = 27 \text{ CV}$

After closing S_2 charge on

$$2C = 2C \cdot \frac{27CV}{3C} = 18 \text{ CV}$$

After closing S_3 charge on

$$4C = 4C \cdot \frac{18CV}{6C} = 12 \text{ CV}$$

After closing S_4 charge on

$$8C = 8C \cdot \frac{12CV}{12C} = 8 \text{ CV}$$

After closing S_5 charge on

$$8C = 8C \cdot \frac{8CV}{24C} = \frac{8CV}{3}$$

Final charge on $8C$ in figure (2)

$$Q = 8C \cdot \frac{\frac{8CV}{3}}{8C + \frac{3C}{2}} = \frac{\frac{64CV}{3}}{\frac{19}{2}} = \frac{128CV}{57}$$

Final charge on capacitor B is

$$Q = \frac{3C}{2} \cdot \frac{\frac{8CV}{3}}{8C + \frac{3C}{2}} = \frac{8CV}{19}$$

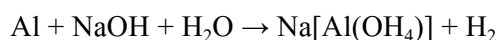
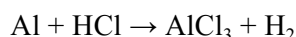
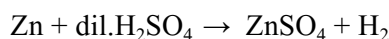
PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (A)**

Deuterium is used for non radioactive labelling and tritium for radioactive labelling

2. **Ans (D)**



3. **Ans (B)**

(A) $\Delta n_g = -1$

$$\frac{4}{3} = K_C \times \left(\frac{1}{12} \times 600 \right)^{-1}$$

$$\frac{4}{3} = \frac{K_C}{5}$$

$$K_C = \frac{20}{3} \text{ Litre mol}^{-1}$$

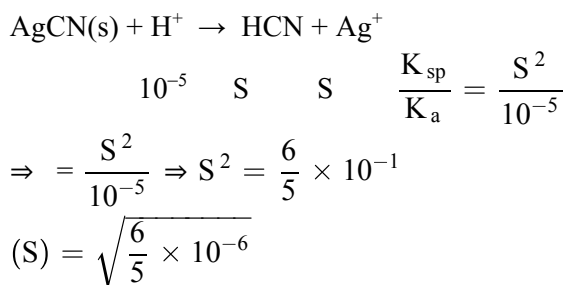
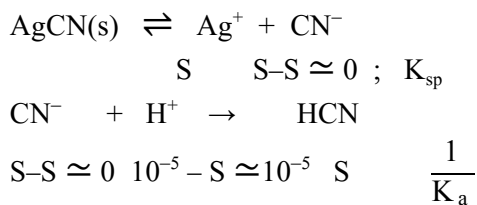
(B) $\Delta n_g = -ve$ means

$$n_{\text{Product}} < n_{\text{Reactants}}$$

So on $\uparrow$ in volume equilibrium shifts in backward direction.

(C) K_p depends on stoichiometry & temperature not on conc.

4. **Ans (B)**



5. **Ans (C)**

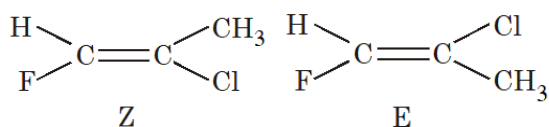
- (a) Trans 1, 2-dimethyl cyclohexane equatorial – equatorial position
(b) Trans 1, 2-dimethyl cyclohexane equatorial – equatorial position
(c) Trans 1, 2-dimethyl cyclohexane axial, axial position.

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (A,C,D)**

Two possible stereoisomers of the given compound are–



In the above shown stereoisomers, I is more polar since, the two dipole vectors C—F and C—Cl are at an angle of 60° and the magnitude of resultant dipole moment will be greater than either of individual bond moment. On the other hand the two dipole vectors, C—F and C—Cl vectors at 180° angle in II and they are in opposite direction, cancelling part of effect of one another, magnitude of resultant dipole vector will be smaller than either of the individual bond moment.

8. **Ans (C)**

Based upon CIP rule

9. **Ans (A,D)**

- (A) In Li & Na $2^{\text{nd}}/3^{\text{rd}}$ P anomaly for EA not applicable.
(B) Li^+ has lower charge as well as higher ionic size.
(D) Theory based.

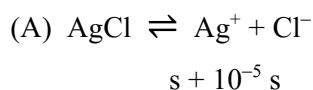
10. **Ans (A)**

Theory based.

11. **Ans (A,C,D)**

As Ni is added, CO will be consumed to attain the third equilibrium. As concentration of CO decreases, COCl_2 will decompose to a higher extent and hence Cl_2 concentration increases which in turn decreases the concentration of PCl_3 .

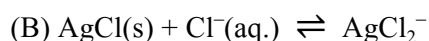
12. **Ans (A,B,C)**



$$s(s + 10^{-5}) = 12 \times 10^{-10}$$

$$\Rightarrow s^2 + 10^{-5} s - 12 \times 10^{-10} = 0$$

$$s = 3 \times 10^{-5} \text{ mole/L}$$



PART-2 : CHEMISTRY

SECTION-II

2. **Ans (250.00)**

$$K_1 = \frac{K_a}{K_w} = \frac{10^{-9}}{10^{-14}} = 10^5$$

$$K_2 = \frac{K_a \times K_b}{K_w} = \frac{10^{-9} \times 10^{-5} \times 2}{10^{-14}} = 2$$

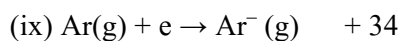
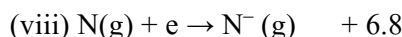
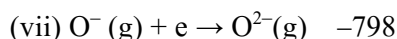
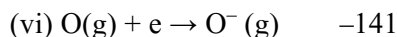
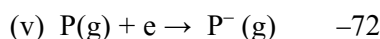
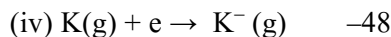
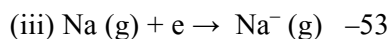
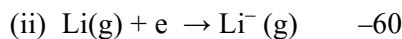
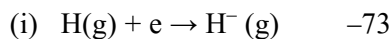
$$K_3 = \frac{10^{-14}}{2 \times 10^{-5}} = \frac{10^{-9}}{2}$$

$$\frac{K_1 \times K_3}{K_2} = \frac{10^5}{2} \times \frac{10^{-9}}{2} = \frac{10^{-4}}{4}$$

$$= \frac{1000 \times 10^{-7}}{4} = 250 \times 10^{-7}$$

3. **Ans (6.00)**

ΔH (KJ/mole)



4. **Ans (5.00)**

(a), (b), (d), (e), & (h) are incorrect.

5. **Ans (3.00)**

Compounds (ii), (iii) and (iv) are optically active.

6. **Ans (420.00)**

4 stereoisomers are possible and all are optically active.

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C)**

$$f'(x) \geq 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow \frac{\sqrt{21-4b-b^2}}{b+1} \leq 1$$

$$21-4b-b^2 \geq 0$$

$$\Rightarrow (b+7)(b-3) \leq 0$$

$$\Rightarrow b \in [-7, 3]$$

$$\text{Case-1 } -7 \leq b < -1$$

$$\text{Case-2 } -1 < b \leq 3$$

$$\Rightarrow 21-4b-b^2 \leq (b+1)^2$$

$$b \leq -5 \text{ or } b \geq 2$$

$$b \in [-7, -1) \cup [2, 3]$$

2. **Ans (A)**

Differentiate the given integral with respect to x

$$u \frac{d^2v}{dx^2} = u \frac{d^2v}{dx^2} + \frac{du}{dx} \cdot \frac{dv}{dx} - v \frac{d^2u}{dx^2} - \frac{dv}{dx} \cdot \frac{du}{dx} + \frac{dw}{dx}$$

$$\Rightarrow \frac{dw}{dx} = v \frac{d^2u}{dx^2}$$

$$\therefore dw = v \frac{d^2u}{dx^2} dx$$

$$\therefore w = \int v \frac{d^2u}{dx^2} dx$$

3. **Ans (C)**

(A) $\cos(\sin x) = \frac{1}{2} \Rightarrow \sin x = \frac{\pi}{3}$ which is not possible.

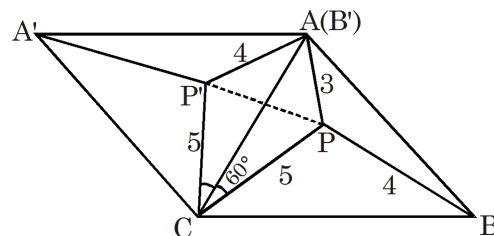
(B) $(1+x^2)^{10} = -[(x-1)^2+1]$ which is not possible.

(C) $L.H.S.]_{\min} = 4; R.H.S.]_{\max} = 4 \Rightarrow x = (2n+1)\pi$

(D) As, $-\sqrt{2} \leq \sin x + \cos x \leq \sqrt{2} \quad \forall x \in \mathbb{R}$

So, $\sin x + \cos x = \frac{\pi}{2}$ (not possible).

4. **Ans (C)**



$\therefore PCP'$ is equilateral

$\therefore PP' = 5$

$\Rightarrow \text{Area of } \triangle PAP' = \frac{1}{2} \times 3 \times 4 = 6$

5. **Ans (B)**

Equation of tangent at $(3\sqrt{3}\cos\theta, \sin\theta)$ is

$$\frac{x \cos \theta}{3\sqrt{3}} + y \sin \theta = 1$$

Its intercept on coordinates axes are $\frac{3\sqrt{3}}{\cos \theta}$ and

$\operatorname{cosec} \theta$. If S denotes their sum, then

$$S = 3\sqrt{3} \sec \theta + \operatorname{cosec} \theta$$

$$\frac{dS}{d\theta} = 3\sqrt{3} \sec \theta \tan \theta - \operatorname{cosec} \theta \cot \theta$$

For minimum value of S, $\frac{dS}{d\theta} = 0$

$$\Rightarrow \frac{3\sqrt{3} \sin \theta}{\cos^2 \theta} - \frac{\cos \theta}{\sin^2 \theta} = 0$$

$$\Rightarrow \tan^3 \theta = \frac{1}{3\sqrt{3}}$$

Since $\theta \in (0, \frac{\pi}{2})$, so $\tan \theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = \frac{\pi}{6}$

Also

$$\frac{d^2S}{d\theta^2} = 3\sqrt{3} \sec \theta \tan^2 \theta + 3\sqrt{3} \sec^3 \theta + 2\cot^2 \theta \operatorname{cosec} \theta + \operatorname{cosec}^3 \theta$$

$$\Rightarrow \left. \frac{d^2S}{d\theta^2} \right|_{\theta=\frac{\pi}{6}} > 0. \text{ Thus } S \text{ is least when } \theta = \frac{\pi}{6}.$$

6. **Ans (D)**

Putting $x = 0$ in the given equation,

We have $f(0) = f'''(3)$ and putting $x = 1$, we get

$$f(1) = 1 + f'(1) + f''(2) + f'''(3).$$

$$\text{Thus, } f(1) - f(0) = 1 + f'(1) + f''(2).$$

Also differentiating the given equation, we have

$$f'(x) = 3x^2 + 2x f'(1) + x f''(2) \quad \dots (i)$$

$$f''(x) = 6x + 2f'(1), f'''(x) = 6$$

$$\text{Thus } f'''(3) = 6 \text{ and } f''(2) = 12 + 2f'(1)$$

Putting $x = 1$ in (i), we have

$$f(1) = 3 + 2f'(1) + f''(2) = 3 + 2f'(1) + 12 + 2$$

$$f(1)$$

$$= 15 + 4f'(1)$$

$$f(1) = -5 \text{ and so } f''(2) = 12 - 10 = 2.$$

$$\Rightarrow f(2) = -2$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (A,B,C)**

$$(A) \quad 2 \cot \frac{B}{2} = \cot \frac{A}{2} + \cot \frac{C}{2}$$

$$\frac{2s(s-b)}{\Delta} = \frac{s(s-a)}{\Delta} + \frac{s(s-c)}{\Delta}$$

$$\Rightarrow 2b = a + c$$

$$(B) \quad \cos A \cot \frac{A}{2} = \left(1 - 2\sin^2 \frac{A}{2}\right) \cot \frac{A}{2}$$

$$= \cot \frac{A}{2} - \sin A$$

$$(C) \quad \frac{s(s-c)}{b} + \frac{s(s-a)}{b} = s = \frac{3b}{2}$$

$$(D) \quad \frac{s(s-a)}{\Delta} \cdot \frac{s(s-c)}{\Delta} = \frac{s}{s-b} = 2$$

8. **Ans (A,B)**

If $y = mx + 5$ is tangent at $P(1, 2)$, so

$$2 = m + 5 \Rightarrow m = \left. \frac{dy}{dx} \right|_{P(1,2)} = -3 \quad \dots (1)$$

$$\text{Now } x^3 y^3 = ax^3 + by^3$$

$\therefore P(1, 2)$ satisfy it

$$\Rightarrow 8 = a + 8b \quad \dots (2)$$

On differentiating with respect to x , we get

$$x^3 \cdot 3y^2 \frac{dy}{dx} + y^3 \cdot 3x^2 = 3ax^2 + 3by^2 \frac{dy}{dx}$$

$$\therefore \text{ Putting } P(x=1, y=2) \text{ and } \left. \frac{dy}{dx} \right|_{P(1,2)} = -3$$

in above equation, we get

$$-4 = a - 12b \quad \dots (3)$$

$$\therefore \text{ On solving (2) and (3), we get } a = \frac{16}{5},$$

$$b = \frac{3}{5}$$

9. Ans (A,B)

$$\frac{\alpha^2}{1 - \tan^2 x} = \frac{(\sin^2 x + \alpha^2 - 2)(1 + \tan^2 x)}{1 - \tan^2 x}$$

$$\Rightarrow \alpha^2 \cos^2 x = \sin^2 x + \alpha^2 - 2$$

$$\Rightarrow 2 = \sin^2 x (1 + \alpha^2)$$

$$\Rightarrow \sin^2 x = \frac{2}{1 + \alpha^2}$$

defined when $x \neq n\pi \pm \frac{\pi}{4}$ and

$$x \neq (2n+1)\frac{\pi}{2}$$

$$\Rightarrow \sin^2 x \in [0, 1) \sim \left\{ \frac{1}{2} \right\}$$

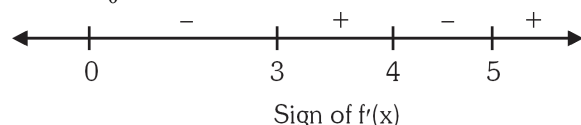
$$0 \leq \frac{2}{1 + \alpha^2} < 1 \text{ and } \frac{2}{1 + \alpha^2} \neq \frac{1}{2}$$

$$\alpha^2 > 1 \text{ and } \alpha^2 \neq 3$$

10. Ans (A,C,D)

We have

$$f(x) = \int_0^x e^{-t^2} (t-5)(t^2 - 7t + 12) dt$$



$$\therefore f'(x) = e^{-x^2} (x-5)(x^2 - 7x + 12)$$

$$= e^{-x^2} (x-3)(x-4)(x-5)$$

Now, verify option (A) and (B)

$$\text{As, } f'(3) = 0 = f'(4) = f'(5)$$

$\therefore$ Consider $y = f'(x)$, on $x \in [3, 4]$

As Rolle's theorem is applicable

So, $\exists$ some $c_1 \in (3, 4)$ such that $f''(c_1) = 0$.

Similarly, consider $y = f'(x)$, on $x \in [4, 5]$.

As Rolle's theorem is applicable

So, $\exists$ some $c_1 \in (4, 5)$ such that $f''(c_2) = 0$.

$\Rightarrow$ There exists atleast two $c_1, c_2 \in (0, \infty)$

such that $f''(c_1) = 0$ and $f''(c_2) = 0$

$\Rightarrow$ Option (C) is correct

Now consider $y = f''(x)$, $x \in [c_1, c_2]$

As Rolle's theorem is applicable

So $f'''(c) = 0$ for some $c \in (0, \infty)$

$\Rightarrow$ Option (D) is correct

11. Ans (B,C)

$$f'(x) = e^{ax}(b\cos(bx+c) + a\sin(bx+c))$$

$$f''(x) = e^{ax}((a^2 - b^2)\sin(bx+c) + 2ab\cos(bx+c))$$

$$= (a^2 + b^2)e^{ax} \sin(bx+c+\phi)$$

$$\text{where } \phi = \tan^{-1} \left(\frac{2ab}{a^2 - b^2} \right)$$

$$= 2\tan^{-1} \left(\frac{b}{a} \right)$$

$$\Rightarrow r^2 = a^2 + b^2 \text{ \& } \theta = c + 2\tan^{-1} \left(\frac{b}{a} \right)$$

12. Ans (A,B,C,D)

(A) $\because f(x)$ is continuous in $[0, 1]$

$\therefore$ Range of $f(x)$ will be $[c, d]$

$$\Rightarrow f(k) \geq f(x) \text{ for one } k \in [0, 1]$$

(B) L.M.V.T. to $f(x)$ in $[0, 1]$

$$f'(c) = \frac{f(1) - f(0)}{1 - 0} \Rightarrow f'(c) = -1$$

where $0 < c < 1$

$f'(x) < 0 \Rightarrow f'(x)$ is decreasing

$$\Rightarrow f'(0) > f'(c) \text{ \& } f'(1) < f'(c)$$

$$\Rightarrow f'(0) > -1 \text{ \& } f'(1) < -1$$

(C) $f(0) = 1$ \& $f(1) = 0$

$$\Rightarrow f(c) = \frac{1+0}{2} \text{ for one } c \in [0, 1]$$

(By I.V.T)

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1.00)

$$\text{Let } f(x) = \sin(\cos(\sin x)) - \cos(\sin(\cos x))$$

$$\Rightarrow f'(x) = -\cos(\cos(\sin x)) \sin(\sin x)(\cos x) - \sin(\sin(\cos x)) \cos(\cos x) \sin x$$

$$\Rightarrow f'(x) < 0 \text{ for all } x \in \left[0, \frac{\pi}{2}\right]$$

$$\Rightarrow f(x) \text{ is decreasing in } \left[0, \frac{\pi}{2}\right]$$

$$\text{and } f(0) = \sin 1 - \cos(\sin 1) \dots (1)$$

$$\text{Now } \sin 1 - \cos(\sin 1)$$

$$= \cos\left(\frac{\pi}{2} - 1\right) - \cos(\sin 1),$$

$$\sin 1 > \sin \frac{\pi}{4} > \frac{1}{\sqrt{2}}$$

$$\therefore \frac{\pi}{2} - 1 < \sin 1$$

$$\Rightarrow \sin 1 > \frac{\pi - 2}{2}$$

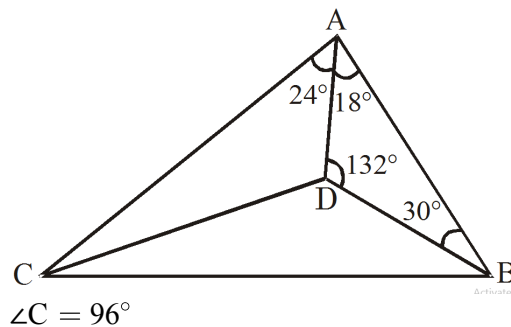
$$\therefore \sin 1 > \cos(\sin 1) \text{ From (1)}$$

$$\therefore f(0) = \sin 1 - \cos(\sin 1) > 0,$$

$$f\left(\frac{\pi}{2}\right) = \sin(\cos 1) - 1 < 0$$

$$\therefore \text{There will be only one root lies between } \left[0, \frac{\pi}{2}\right].$$

2. Ans (4.00)



$$\therefore A = B = \frac{180^\circ - 96^\circ}{2} = 42^\circ$$

$$\Delta ABC, \frac{AB}{\sin 96^\circ} = \frac{AC}{\sin 42^\circ}$$

$$AB = AC \frac{2 \sin 48^\circ \cos 48^\circ}{\cos 48^\circ} = 2AC \sin 48^\circ \dots (i)$$

$$\text{In } \Delta ABD, \frac{AB}{\sin 132^\circ} = \frac{AD}{\sin 30^\circ}$$

$$\Rightarrow AB = 2AD \sin 132^\circ = 2AD \sin 48^\circ \dots (ii)$$

$$(i) + (ii) \Rightarrow AC = AD$$

$$\therefore \angle ADC = \frac{180^\circ - 24^\circ}{2} = 78^\circ$$

3. Ans (1.00)

$$\therefore \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$$

$$\begin{aligned} \frac{d^2x}{dy^2} &= \frac{d}{dy} \left(\frac{1}{\frac{dy}{dx}} \right) = \frac{d}{dx} \left(\frac{1}{\frac{dy}{dx}} \right) \cdot \frac{dx}{dy} \\ &= \frac{-1}{\left(\frac{dy}{dx}\right)^2} \cdot \frac{d^2y}{dx^2} \cdot \frac{dx}{dy} = \frac{-1}{\left(\frac{dy}{dx}\right)^3} \cdot \frac{d^2y}{dx^2} \end{aligned}$$

$$\text{From the given is } \frac{d^2y}{dx^2} \left(1 - \frac{1}{\left(\frac{dy}{dx}\right)^3} \right) = 0$$

$$\Rightarrow \frac{d^2y}{dx^2} = 0 \text{ or } \frac{dy}{dx} = 1 \Rightarrow y = x + 1$$

4. **Ans (1.50)**

$$e^{3x} = k^x \Rightarrow x = 0$$

$$\Rightarrow m_1 = 3e^{3x} \Big|_{x=0} = 3 \text{ \&}$$

$$m_2 = k^x \ln k \Big|_{x=0} = \ln k$$

$$\tan 45^\circ = \left| \frac{3 - \ln k}{1 + 3 \ln k} \right|$$

$$\Rightarrow \ln k = \frac{1}{2} \text{ or } \ln k = -2$$

$$\Rightarrow \text{Product} = e^{\frac{1}{2} - 2} = e^{-3/2}$$

5. **Ans (8.00)**

$$f(3) = 5 \text{ \& } f'(3) = \frac{5}{4}$$

$$\lim_{x \rightarrow 3} \frac{(3^{4f(x)} - 3^{20})^2}{1 - \cos(\ln(4-x))}$$

$$= \lim_{x \rightarrow 3} \frac{3^{40} \left(\frac{3^{4(f(x)-5)} - 1}{(4f(x)-20)} \right)^2 16(f(x)-5)^2}{\frac{1 - \cos(\ln(4-x))}{(\ln(4-x))^2} \left(\frac{\ln(4-x)}{3-x} \right)^2 (3-x)^2}$$

$$= 2 \times 3^{40} (\ln 3)^2 \times 25$$

6. **Ans (43.00)**

Since $f(x)$ has its domain as $\mathbb{R}$

$$\therefore a - 3 = 0 \Rightarrow a = 3$$

$$f(x) = \frac{bx}{x^2 + 4} = y$$

$$bx = x^2y + 4y \Rightarrow x^2y - bx + 4y = 0$$

$$b^2 - 16y^2 \geq 0 \Rightarrow y \in \left[-\frac{b}{4}, \frac{b}{4} \right]$$

$$\Rightarrow \frac{b}{4} = 8 \Rightarrow b = 32$$

$$\text{Now } f(x) = \frac{32x}{x^2 + 4} = mx$$

$$\Rightarrow x = 0, x^2 = \frac{32}{m} - 4$$

$$\text{for three solutions } \frac{32}{m} - 4 > 0$$

$$\Rightarrow \frac{m-8}{m} < 0$$

$$\Rightarrow m \in (0, 8)$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

PART TEST

28-01-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I & I(N)] : (SCORE-I)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	A	C	D	A	D	B	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	A	C	C	A	C	A	A	C	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	14.00	251.00	1.00	2.00	37.50	2.60	2.50	54.00	8.00	594.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	C	C	B	C	B	B	A	C
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	C	B	B	B	D	D	B	A	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	1.00	4.00	1.00	4.00	3.47	5.00	82.00	2.00	51.00	9.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	A	A	B	D	A	D	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	A	C	B	C	B	C	C	A	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	63.50	6.00	20.00	4.00	-0.50	1.00	20.00	3.00	5050.00	2.00

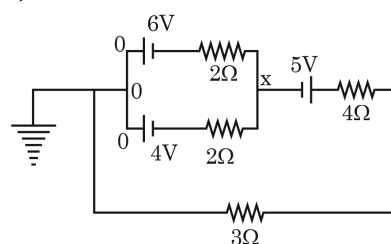
HINT – SHEET
PART-1 : PHYSICS
SECTION-I
1. Ans (B)

$$R_{eq} = \int_0^L \frac{\rho dx}{\pi \left(\frac{a+(b-a)x}{L} \right)^2}$$

$$= \frac{\rho L}{\pi ab}$$

$$E = \frac{\rho i}{A} = \rho \times \frac{V}{\frac{\rho L}{\pi ab} \times \pi \left(a + (b-a) \frac{x}{L} \right)^2}$$

$$E = \frac{V ab}{L \left(a + (b-a) \frac{x}{L} \right)^2}$$

2. Ans (D)

Node method

$$\frac{x+6}{2} + \frac{x+4}{2} + \frac{x+5}{7} = 0$$

$$x = -5$$

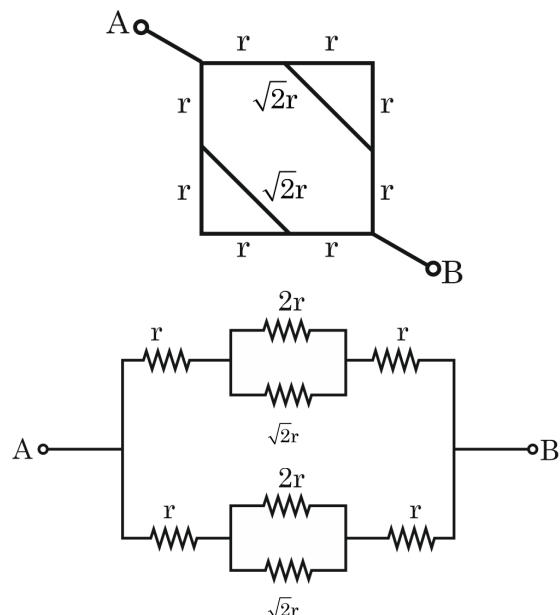
 so current in $4\Omega = 0$

$$V_{AB} = 0$$

3. Ans (A)

The circuit is equivalent to

Let each half side has resistance $r = \rho d/2$.

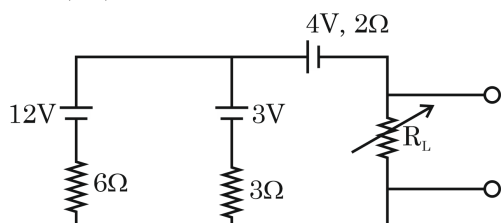


On solving, we get

$$R = \frac{1}{2} \left[2r + \frac{(2r)(r\sqrt{2})}{(2 + \sqrt{2})r} \right] = r\sqrt{2}$$

$$= \frac{\rho d}{\sqrt{2}}$$

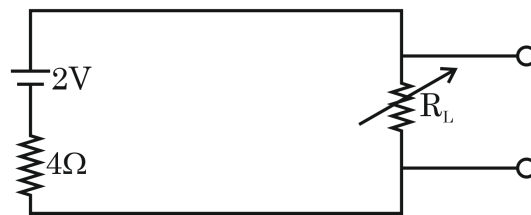
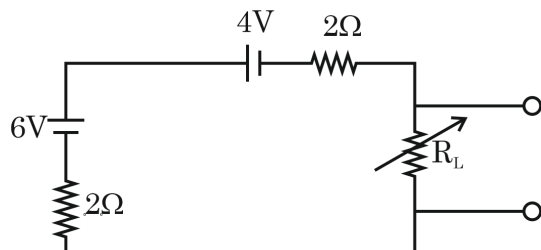
4. Ans (C)



12V & 3V are in parallel

$$\frac{\frac{12}{6} + \frac{3}{3}}{\frac{1}{6} + \frac{1}{3}} = \varepsilon$$

$$\varepsilon = 6$$



for $P_{\max}$

$$R_L = 4\Omega$$

$$i = \frac{2}{4 + 4} = \frac{2}{8} = \frac{1}{4}$$

$$P_{\max} = \left(\frac{1}{4} \right)^2 \times 4$$

$$= 0.25 \text{ Watt}$$

5. Ans (D)

when $I = 0$

$x = 20 \text{ cm}$

$$\text{It means } V_{AC} = \frac{V_0 x}{L} = \frac{12 \times 20}{100} = 2.4 \text{ volt}$$

Since there is no current through the milliammeter

$$\text{Hence } V_{AC} = V = 2.4 \text{ volt}$$

When $x = 0$; $I = 30 \text{ mA}$

$$\therefore 2.4 = IR$$

$$2.4 = 30 \times 10^{-3} \cdot R$$

$$R = 80\Omega$$

6. Ans (A)

$$\text{Least count} = \frac{1 \text{ mm}}{50}$$

$$\text{Zero error} = -10 \times \frac{1}{50} \text{ mm} = -0.2 \text{ mm}$$

$$\text{Reading} = 3 \text{ mm} + 25 \times \frac{1}{50} \text{ mm} - (-0.2 \text{ mm})$$

$$\text{Reading} = 3.7 \text{ mm}$$

7. Ans (D)

$$8 \text{ VSD} = 10 \text{ MSD}$$

$$\text{VSD} = \frac{10 \times 1 \text{ mm}}{8}$$

$$\text{VSD} = \frac{10}{8} = 1.25 \text{ mm}$$

$$= 35 \text{ MSD} - 6 \text{ VSD (according to diagram)}$$

$$= 2.75 \text{ cm}$$

8. **Ans (B)**

$$V = 2f_0 (l_2 - l_1)$$

$$(dV) = 2f_0 (dl_2 - dl_1)$$

$$(dV)_{\max} = \max \text{ of } [2f_0 (\pm \Delta l_2 \pm \Delta l_1)]$$

$$= 2f_0 (\Delta l_2 + \Delta l_1)$$

$$l_1 = 25.0 \text{ cm} \Rightarrow \Delta l_1 = 0.1 \text{ cm (place value of last number)}$$

$$l_2 = 75.0 \text{ cm} \Rightarrow \Delta l_2 = 0.1 \text{ cm (place value of last number)}$$

So the maximum permissible error in the speed of sound $(dV)_{\max}$

$$= 2(325\text{Hz})(0.1 \text{ cm} + 0.1 \text{ cm}) = 1.3 \text{ m s}^{-1}$$

9. **Ans (B)**

$$34.123 + 2.10 = 36.223 = 36.22$$

$$98.7654 - 4.983 = 93.7824 = 93.782$$

$$\frac{36.22}{93.782} = 0.386214... = 0.3862$$

(lowest significant value)

10. **Ans (B)**

$$\frac{R_x}{R} = \frac{x}{L-x}$$

$$R_x = R \left(\frac{x}{L-x} \right)$$

$$\text{For } R = 20\Omega, x = 20 \text{ cm}$$

$$R_x = 20 \left[\frac{20}{100-20} \right] = 5\Omega$$

$$\text{from (i) } \ln R_x = \ln r + \ln x - \ln (L-x)$$

$$\frac{dR_x}{R_x} = \frac{dx}{x} + \frac{dx}{L-x}$$

$$\frac{\Delta R_x}{R_x} = \frac{L\Delta x}{x(L-x)} \quad [\Delta x = \text{maximum error in } x]$$

$$\text{for } R_x = 5\Omega; |\Delta x| = 0.1 \text{ cm};$$

$$L = 100 \text{ cm } x = 20 \text{ cm}$$

$$\therefore |\Delta R_x| = \frac{5 \times 100 \times 0.1}{20 \times 80} = 0.03\Omega$$

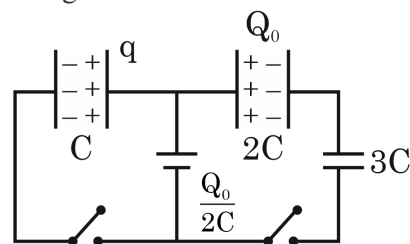
$$\therefore R_x = (5.00 \pm 0.03)\Omega$$

11. **Ans (C)**

$$= \frac{2\epsilon_0 A}{5d} + \frac{2\epsilon_0 A}{5\left(\frac{3d}{4}\right)} + \frac{2\epsilon_0 A}{5(d)} = \frac{4\epsilon_0 A}{3d}$$

12. **Ans (A)**

After closing the switches.



$$q = \frac{Q_0}{2C} \times C$$

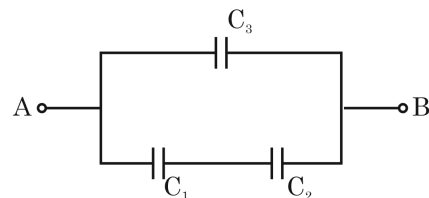
$$\frac{Q_0}{2}$$

$\therefore$ charge flown through

$$S_2 = \frac{Q_0}{2} - \frac{Q_0}{3} = \frac{Q_0}{6}$$

charge flown through $S_1 = 0$

13. **Ans (C)**



$$C = \frac{\epsilon_0 A}{d} \text{ assume}$$

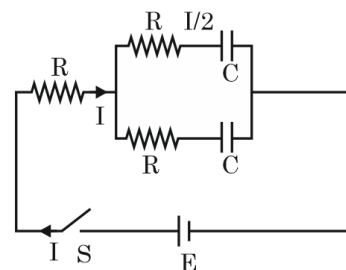
$$C_1 = \frac{2\epsilon_0 A}{d} = 2C$$

$$C_2 = \frac{3\epsilon_0 A}{d} = 3C$$

$$C_3 = \frac{\epsilon_0 A}{2d} = \frac{C}{2}$$

$$C_{AB} = \frac{C_1 C_2}{C_1 + C_2} + C_3 = \frac{17\epsilon_0 A}{10d}$$

14. **Ans (C)**



Applying KVL, we get

$$\frac{I}{2} = \frac{dq}{dt} \text{ or } I = 2 \frac{dq}{dt}$$

$$E - IR - \frac{I}{2}R - \frac{q}{C} = 0$$

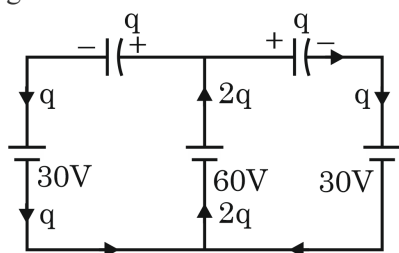
$$\text{or } E - \frac{3}{2}IR - \frac{q}{C} = 0$$

$$\text{or } E - \frac{3}{2} \times 2 \frac{dq}{dt} R - \frac{q}{C} = 0$$

$$q(t) = EC (1 - e^{-t/3RC})$$

15. Ans (A)

With switch S open the potential difference across the group of capacitors is zero. There is no charge on both of them.



With S close, potential difference across both of them is 30V. Charge on each of them is

$$q = (2 \mu\text{F}) (30 \text{ V}) = 60 \mu\text{C}$$

Work done by 60V battery

$$= 2q \times 60 = 2 \times 60 \times 60 = 7200 \mu\text{J}$$

Work done by 30V battery (each)

$$= -q \times 30 = -60 \times 30 = -1800 \mu\text{J}$$

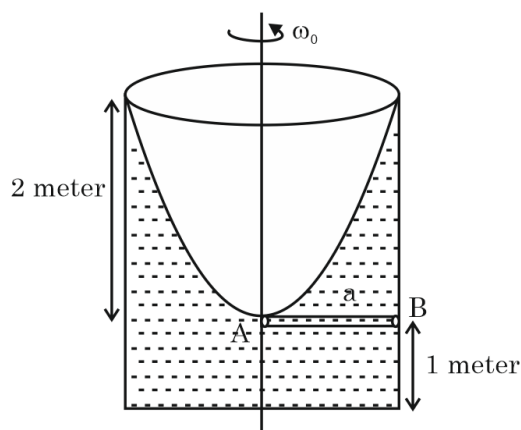
Work done by battery = $\Delta U + HL$

$$7200 \mu\text{J} + (-3600 \mu\text{J}) = \frac{1}{2} \frac{q^2}{C} \times 2 + HL$$

put q and C

$$HL = 1.8 \text{ mJ}$$

16. Ans (C)



centrifugal acceleration

$$a_c = \frac{R}{2} \omega^2 = r_{cm} \omega^2$$

$$P_B = P_A + \rho a_c R$$

$$P_0 + \rho g \times 2 = P_0 + \frac{\rho R}{2} \omega^2$$

$$\omega = \sqrt{40}$$

17. Ans (A)

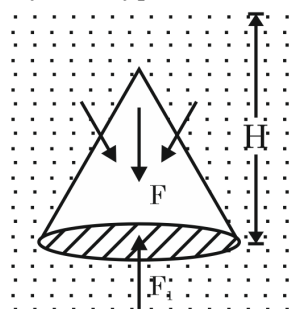
Buoyancy force is $F_B = \rho V g$

Weight of the cone $V = \rho V g$

$$\therefore \text{Tension } T = F_B - mg = \rho V g (\rho - d)$$

$$F_B = \rho V g$$

Buoyancy is resultant of vertically upward force (F_1) applied by water pressure on the base of the cone and the vertically downward force (F) applied by water on the slant surface. [The horizontal component of this force sums upto zero due to symmetry]



$$\therefore F_1 - F = F_B$$

$$\pi R^2 [P_0 + \rho g H] - F = \rho V g$$

$$\therefore F = \pi R^2 [P_0 + \rho g H] - \rho V g$$

18. Ans (A)

Volume of the portion is the plate immersed in water is

$$10 \times \frac{1}{2} (1.54) \times 0.2 = 1.54 \text{ cm}^3$$

Therefore, if the density of water is taken as 1, then upthrust

= weight of the water displaced

$$= 1.54 \times 1 \times 980 = 1509.2 \text{ dynes}$$

Now, the total length of the plate in contact with the water surface is $2(10 + 0.2) = 20.4 \text{ cm}$

$\therefore$ Downward pull upon the plate due to surface tension

$$= 20.4 \times 73 = 1489.2 \text{ dynes}$$

$\therefore$ Resultant upthrust

$$= 1509.2 - 1489.2$$

$$= 20.0 \text{ dynes} = \frac{20}{980}$$

$$= 0.0204 \text{ gm - wt}$$

$\therefore$ Apparent weight of the plate in water

= weight of the plate in water

$$= 8.2 - 0.0204 = 8.1796 \text{ gm-wt}$$

19. Ans (C)

$$\rho = Fv$$

$$= 6\pi\eta rv^2$$

$$= 6\pi\eta r \times \frac{GM}{R} = 3 \times 10^6 W$$

20. Ans (A)

If V_2 is speed at point 2.

$$A_2 V_2 = A_3 \times 6$$

$$\Rightarrow V_2 = 6.75 \text{ m/s}$$

Applying Bernoulli's equation for point 2 and 3

$$P_2 + \frac{1}{2}\rho_{Hg} V_2^2 = P_{atm} + \frac{1}{2}\rho_{Hg} \cdot V^2$$

$$P_2 = 1.01 \times 10^5 + \frac{1}{2} \times 1.36 \times 10^4 (6^2 - 6.75^2)$$

$$= 0.36 \times 10^5 \text{ N/m}^2$$

$$\therefore \rho_{Hg}gh = 0.36 \times 10^5$$

$$h = \frac{0.36 \times 10^5}{1.36 \times 10^4 \times 10} = 26 \text{ cm}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (14.00)

From the table we can see that

Brown colour $\rightarrow 1$

Yellow colour $\rightarrow 4$

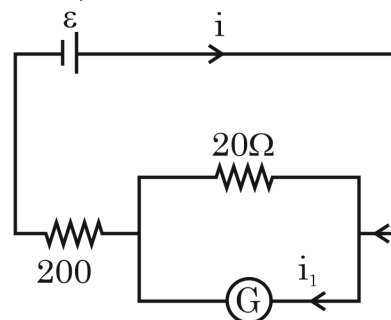
Green colour $\rightarrow 10^5$ and

Gold colour $\rightarrow 5\%$

$$R = (14 \times 10^5 \pm 5\%) \Omega$$

$$(a = 14, b = 5, n = 5)$$

2. Ans (251.00)

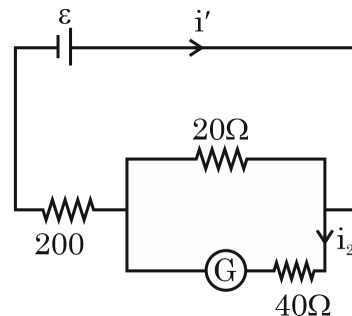


$$i = \frac{\varepsilon}{200 + \frac{20\Omega}{20+G}}$$

$$i_1 = \frac{20i}{20 + G}$$

$$i_1 = \frac{20}{(20 + G)} \frac{\varepsilon}{\left(200 + \frac{20G}{(20+G)}\right)}$$

$$i_1 = \frac{20\varepsilon}{200(20 + G) + 20G}$$



$$i' = \frac{\varepsilon}{200 + \frac{20(G+40)}{(G+60)}}$$

$$i_2 = \frac{i'20}{(60 + G)}$$

put i'

$$i_2 = \frac{\varepsilon 20}{200(60 + G) + 20G + 800}$$

$$i_2 = \frac{i_1}{2}$$

$$\frac{\varepsilon 20}{12000 + 200G + 20G + 800} = \frac{\varepsilon 20}{2[4000 + 200G + 20G]}$$

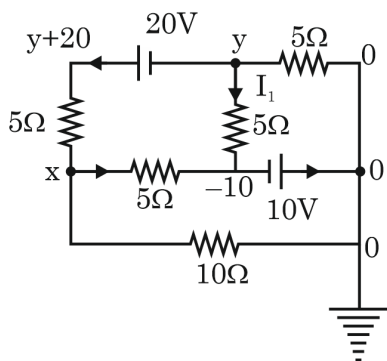
$$12800 + 220G = 8000 + 440G$$

$$4800 = 220G$$

$$G = \frac{240}{11}$$

$$a = 240, b = 11 \Rightarrow a + b = 251$$

3. Ans (1.00)



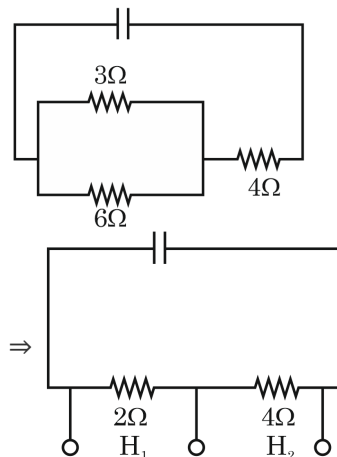
$$\frac{x}{10} + \frac{x+10}{5} + \frac{x-y-20}{5} = 0$$

$$\frac{y}{5} + \frac{y+10}{5} + \frac{y+20-x}{5} = 0$$

On solving for x and y we get $x = 0$, $y = -10$.

Hence $I_1 = 0$, $I_2 = 2$ and $I_3 = 2$ A, i.e., I_4 will also be 2 A.

4. Ans (2.00)



$$\text{Total heat} = \frac{1}{2} \frac{q^2}{C} = H_0$$

in series of 2Ω & 4Ω current will be same so

$$H \propto R$$

$$\frac{H_1}{H_2} = \frac{2}{4}$$

$$2H_1 = H_2$$

$$H_1 + H_2 = H_0$$

$$H_1 + 2H_1 = H_0$$

$$H_1 = \frac{H_0}{3}$$

H_1 will be across 3Ω & 6Ω both are in parallel

$$\text{Heat across } 3\Omega = H_3$$

$$\text{Heat across } 6\Omega = H_4$$

$$H_3 + H_4 = H_1 = \frac{H_0}{3}$$

3Ω & 6Ω

are in parallel

$$\text{so } H \propto \frac{1}{R}$$

$$\frac{H_3}{H_4} = \frac{6}{3}$$

$$H_3 = 2H_4$$

$$H_3 + H_4 = \frac{H_0}{3}$$

$$2H_4 + H_4 = \frac{H_0}{3}$$

$$H_4 = \frac{H_0}{9}$$

$$H_0 = \frac{(6)^2}{2 \times 1 \times 9} = 2\mu\text{J}$$

5. Ans (37.50)

$$\text{Force on dielectrislab} = F = \frac{1}{2} \frac{\epsilon_0 b}{d} V^2 (k - 1)$$

block is moving constant velocity

$$m'g = F$$

$$m'g = \frac{1}{2} \frac{\epsilon_0 b}{d} V^2 (k - 1)$$

$$m'g = \frac{1}{2} \times 10 \times (5)^2 (4 - 1)$$

$$m' = \frac{25}{2} \times 3 = 37.5 \text{ kg}$$

6. Ans (2.60)

$$\text{Zero error} = 10 \times \frac{1}{100} = 0.1 \text{ mm}$$

$$\text{diameter} = 2\text{mm} + \frac{1}{100} \times 70 - 0.1$$

$$= 2.6 \text{ mm}$$

7. Ans (2.50)

$$W = A^3 \ln(at^2)$$

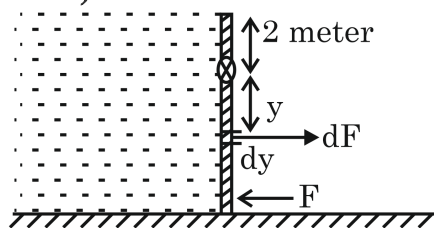
$$dW = 3A^2 dA \ln(at^2) + \frac{A^3}{at^2} \times 2at dt$$

$$dW = 3A^2 dA \ln(at^2) + \frac{A^3 2dt}{t}$$

$$\frac{dW}{W} = \frac{3A^2 dA}{A^3} + \frac{2dt}{t \ln(at^2)}$$

$$\frac{dW}{W} \times 100 = \frac{3dA}{A} \times 100 + \frac{2}{\ln(12)} \left(\frac{dt}{t} \right) \times 100$$

8. Ans (54.00)



$$\text{Torque of water on the gate} = \int y dF$$

$$= \int y \rho g (2 + y) dy$$

$$= 6 \int_{-2}^4 \rho g (2y + y^2) dy$$

$$= 6 \rho g \left[\frac{2y^2}{2} + \frac{y^3}{3} \right]_{-2}^4$$

$$\tau_{\text{water}} = 2 \rho g (108)$$

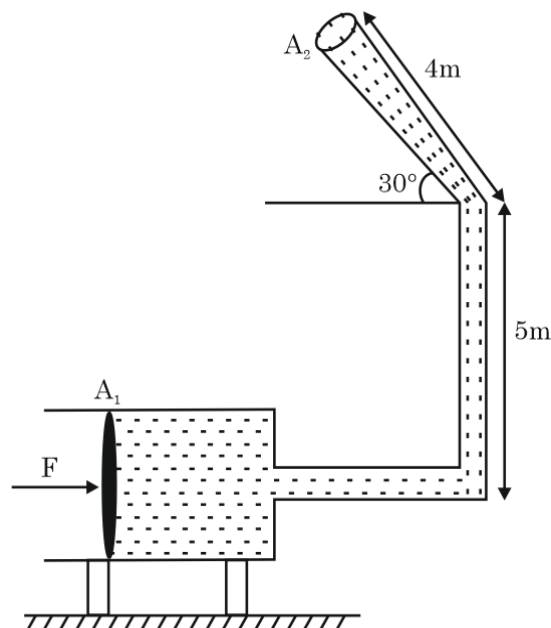
$$\tau_{\text{net}} = 0$$

$$\text{Torque of water} = \text{Torque of } F$$

$$2 \rho g (108) = 4F$$

$$F = 54 \rho g$$

9. Ans (8.00)



Conservation of energy

$$\frac{100}{A_1} + P_0 + \frac{1}{2} \rho V_1^2 = P_0 + \frac{1}{2} \rho V_2^2 + \rho g$$

$$(H_1 + L \sin 30^\circ)$$

$$\frac{100}{A_1} + \frac{1}{2} \rho V_1^2 = \frac{1}{2} \rho V_2^2 + \rho g (H_1 + L \sin 30^\circ)$$

$$A_1 V_1 = A_2 V_2$$

$$4V_1 = V_2$$

$$V_1 = \frac{V_2}{4}$$

$$\frac{100}{A_1} - \rho g \left(H_1 + \frac{L}{2} \right) = \frac{1}{2} \rho \left(V_2^2 - \frac{V_2^2}{16} \right)$$

$$\frac{100 \times 2}{10 \times 10^{-4} \times 100} - 20 \left(H_1 + \frac{L}{2} \right) = \frac{15}{16} V_2^2$$

$$200 - 20 \left[5 + \frac{4}{2} \right] = \frac{15}{16} V_2^2$$

$$60 = \frac{15}{16} V_2^2$$

$$\sqrt{4 \times 16} = V_2$$

$$V_2 = 2 \times 4 = 8 \text{ m/s}$$

10. **Ans (594.00)**

Initial acceleration for larger ball is given by -

$$ma = mg - F_B$$

$$V_1 d_1 \frac{g}{2} = V_1 d_1 g \times V_1 \rho g$$

[ρ = density of water, d_1 = density of ball]

$$\Rightarrow d_1 = 2\rho$$

$$\text{Similarly, for second ball } d_2 = \frac{4\rho}{3}$$

After long time both of them will acquire terminal speed (u_{01} for larger ball and u_{02} for smaller ball)

$$6\pi\eta r u_{01} = mg - F_B$$

$$\Rightarrow 6\pi\eta r u_{01} = \frac{4}{3} \pi r^3 d_1 g - \frac{4}{3} \pi r^3 \rho g$$

$$\Rightarrow u_{01} = \frac{2}{9} \frac{r^2 \rho g}{\eta} \quad \text{similarly, } u_{02} = \frac{r^2 \rho g}{54\eta}$$

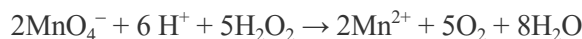
$\therefore$ Velocity of smaller ball with respect to the larger ball is

$$= \left(\frac{1}{54} - \frac{2}{9} \right) r^2 \frac{\rho g}{\eta} = -\frac{11}{54} \frac{r^2 \rho g}{\eta} = \frac{11}{54} \frac{r^2 \rho g}{\eta} (\uparrow)$$

PART-2 : CHEMISTRY

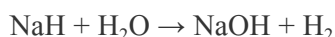
SECTION-I

1. **Ans (A)**



2. **Ans (D)**

Very pure hydrogen (99.9%) is prepared by the action of water on salt hydrides (eg. NaH)



3. **Ans (C)**

Organic waste consumes oxygen and therefore, dissolved oxygen in water decreases and fish in such polluted water die.

4. **Ans (C)**

Due to water pollution solubility of oxygen in water decreases which make oxygen molecules unavailable to water plants.

5. **Ans (B)**

Ununtrium for atomic number

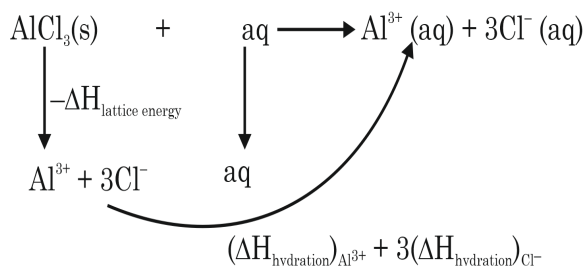
$$= 113; \text{ for } Z > 86$$

period number = 7

$$113 - [\text{Rn}] 7s^2, 5f^{14}, 6d^{10}, 7p^1$$

hence Ununtrium is a p-block element.

6. **Ans (C)**



$$\Rightarrow \Delta H_{\text{solution}} = -4665 - 3 \times 381 + 5137 < 0$$

Hence AlCl_3 will dissolve and solution consists of hydrated Al^{3+} and Cl^- ions. In other words higher hydration energy compensates the higher ionisation energy. As a result, it will exist as hydrated Al^{3+} and hydrated Cl^- in aqueous solution.

7. **Ans (B)**

The second ionisation potential of Mg is lesser than the second ionisation potential of Na. The removal of second electron from Mg takes place from 3s-orbital whereas the removal of second electron from N a takes place from 2p-orbital. More closer are shells to the nucleus, difficult is removal of electron.

8. Ans (B)

$\text{LaCl}_3(\text{s}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{LaClO}(\text{s}) + 2\text{HCl}(\text{g})$ -
 heat already at equilibrium, more water vapour
 is added without altering T or V of the system.
 When equilibrium re-established, the pressure
 of water vapour is doubled. The pressure of
 HCl present in the system increases by a factor
 of $2^{1/2}$.

The equilibrium constant expression is

$$K = \frac{[\text{HCl}]^2}{[\text{H}_2\text{O}]}$$

Before addition of water vapour,

$$K = \frac{[\text{HCl}]_i^2}{[\text{H}_2\text{O}]_i} \dots(1)$$

After addition of water vapour,

$$K = \frac{[\text{HCl}]_f^2}{[\text{H}_2\text{O}]_f} \dots(2)$$

But (1) = (2)

$$\frac{[\text{HCl}]_i^2}{[\text{H}_2\text{O}]_i} = \frac{[\text{HCl}]_f^2}{[\text{H}_2\text{O}]_f} \dots(3)$$

When equilibrium re-established, the pressure
 of water vapour is doubled.

$$[\text{H}_2\text{O}]_f = 2[\text{H}_2\text{O}]_i \dots(4)$$

Substitute (4) in (3)

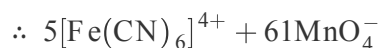
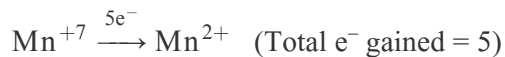
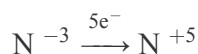
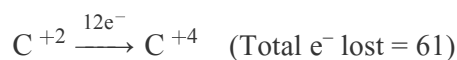
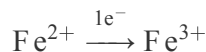
$$\frac{[\text{HCl}]_i^2}{[\text{H}_2\text{O}]_i} = \frac{[\text{HCl}]_f^2}{(2[\text{H}_2\text{O}]_i)}$$

$$[\text{HCl}]_i^2 = [\text{HCl}]_f^2 \times \frac{1}{2}$$

$$[\text{HCl}]_i = [\text{HCl}]_f \times \sqrt{\frac{1}{2}}$$

$$[\text{HCl}]_f = \sqrt{2} \times [\text{HCl}]_i$$

9. Ans (A)



10. Ans (C)

$$[\text{S}^{2-}] = \frac{10^{-17}}{0.1} = 10^{-16}$$

$$K = 10^{-21} = \frac{[\text{H}^+]^2 \times 10^{-16}}{0.1}$$

$$10^{-6} = [\text{H}^+]^2$$

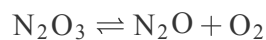
$$[\text{H}^+] = 10^{-3}$$

$$\therefore \text{pH} = 3$$

11. Ans (D)



Moles at equilibrium $4 - x \quad x - y \quad x + y$



mole at equilibrium $x - y \quad y \quad x + y$

$$K_C = 4.5 = \frac{(x + y)(x - y)}{4 - x} \text{ and } x + y = 4.5$$

$$\Rightarrow x = \frac{17}{6} \text{ and } y = \frac{5}{3}$$

K_C for the dissociation of N_2O_3 can be
 calculated as follows:

$$K_C = \frac{4.5 \times 5 \times 6}{3 \times 7} = 6.428$$

12. Ans (C)

$$[\text{H}^+] - [\text{OH}^-] = C_{\text{HCl}}$$

$$[\text{H}^+] = C_{\text{HCl}} + \frac{K_W}{[\text{H}^+]}$$

13. Ans (B)

In presence of phenolphthalein,

$$\frac{1}{2} \text{ meq. of Na}_2\text{CO}_3 = \text{meq of H}_2\text{SO}_4$$

$$\frac{1}{2} N_{\text{Na}_2\text{CO}_3} \times 10 = 2 \times 0.1 \times 2.5$$

$$N_{(\text{Na}_2\text{CO}_3)} = 0.1$$

$$\therefore \text{amount of Na}_2\text{CO}_3 \text{ in g/L} = N \times \text{Eq. wt of Na}_2\text{CO}_3$$

$$= 0.1 \times 53$$

$$= 5.3 \text{ g}$$

The amount of NaHCO_3 in g/L = 4.2

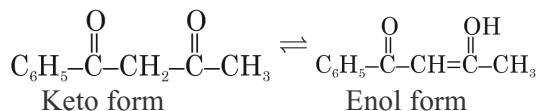
14. Ans (B)

Volume of solid is more than volume of liquid water. Hence increase in pressure moves the equilibrium forward.

15. Ans (B)

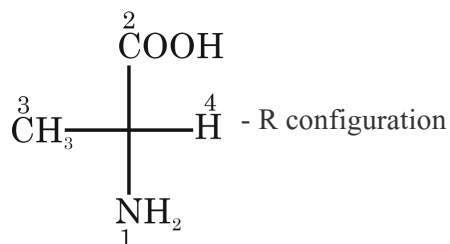
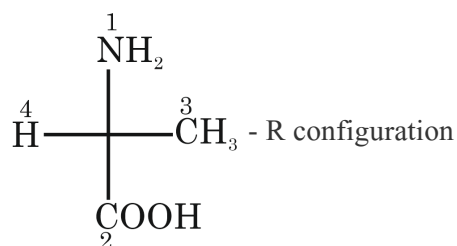
Compound is dibromoethene, it can have 3 isomers (two geometrical isomers (cis & trans) of 1,2 dibromoethene and one is 1,1 dibromoethene).

16. Ans (D)



While in option A, B, C there is no acidic hydrogen so these compound will no show tautomerism.

17. Ans (D)



18. Ans (B)

(A) has plane of symmetry so it is optically inactive

(B) it has a chiral center so it is optically active

(C) has not any chiral center so it is optically inactive

(D) due to amine inversion so it is optically inactive

19. Ans (A)

Let s mol/litre be the solubility of $M(OH)_2$

$$s = [M^{2+}] + [M(OH)_4]^{2-}$$

$$= \frac{K_{sp}}{[\text{OH}^-]^2} + K_f[\text{OH}^-]^2$$

$$\frac{ds}{d[\text{OH}^-]} = \frac{-2K_{\text{sp}}}{[\text{OH}^-]^3} + 2K_{\text{f}}[\text{OH}^-]$$

for minimum solubility

$$\frac{ds}{d[\text{OH}^-]} = 0$$

$$\frac{K_{sp}}{K_f} = [\text{OH}^-]^4$$

$$10^{-16} = [\text{OH}^-]^4$$

$$[\text{OH}^-] = 10^{-4}$$

20. Ans (A)

4 geometrical isomers which are (trans, trans), (trans, cis), (cis, cis), (cis, trans) and each geometrical isomers will have 2 optical isomers. So, a total of 8 stereoisomers will be obtained.

PART-2 : CHEMISTRY

SECTION-II

1. Ans (1.00)

$$S = \frac{K_{eq} P_{O_2}}{1 + K_{eq} P_{O_2}}$$

$$0.5 = \frac{K_{eq} \times 3}{1 + 3K_{eq}}$$

$$\Rightarrow K_{eq} = 1/3 \text{ atm}^{-1}$$

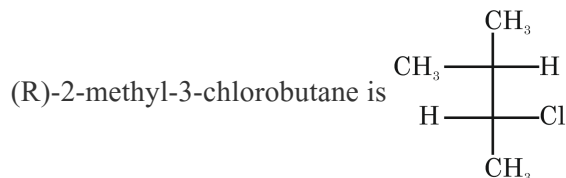
2. **Ans (4.00)**

4 geometrical isomers namely: (cis, cis), (cis, trans), (trans, cis) and (trans, trans).

3. **Ans (1.00)**

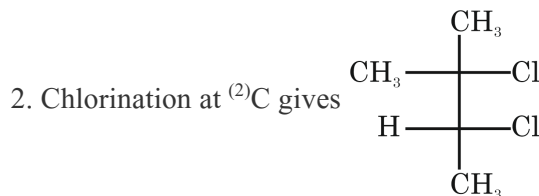
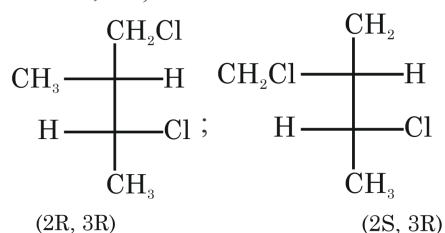
In option (d) there is a plane of symmetry in the compound.

4. **Ans (4.00)**



Four structural isomers are as follows.

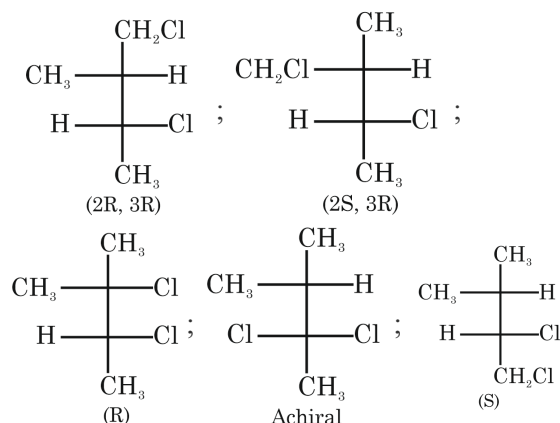
1. Chlorination at ^1C creates a new chiral centre at ^2C . Since this C can be R or S, two diastereomers are formed. (These are 2R, 3R and 2S, 3R)



3. Chlorination at ^3C result in the loss of chirality.

4. Chlorination at ^4C results into S as the priority sequence is changed, though its configuration remains the same.

Thus, The possible products are



Thus, four stereoisomers are possible.

5. **Ans (3.47)**

For 1 mole

$$\Delta H = 19\text{K Cal}$$

$$= 19 \times 2.611 \times 10^{22} = 4.96 \times 10^{23} \text{ eV/mol}$$

for 1 atom,

$$\Delta H = \frac{4.96 \times 10^{23}}{6.023 \times 10^{23}} = 0.823 \text{ eV/atom}$$

$$\text{Sp, } e^- \text{ affinity of F} = (4.3 - 0.823) \text{ eV} = 3.477 \text{ eV}$$

6. **Ans (5.00)**

$$\text{EN of B} = 2.0$$

$$\text{EN of Si} = 1.8$$

$$\text{EN of Mg} = 1.2$$

$$= 5.0$$

8. **Ans (2.00)**

Only NH_4Cl reacts with NaOH

$$\begin{aligned} \text{meq of NaOH left} &= \text{meq of HCl used} \\ &= 52.4 \times 0.1 \\ &= 5.24 \end{aligned}$$

$$\begin{aligned} \therefore \text{meq of NH}_4\text{Cl} &= \text{meq of NaOH used for NH}_4\text{Cl} \\ &= (100 \times 0.8) - 5.24 \\ &= 74.76 \end{aligned}$$

$$\frac{W_{\text{NH}_4\text{Cl}}}{53.5} \times 100 = 74.76$$

$$W_{\text{NH}_4\text{Cl}} = 4\text{g}$$

$$\therefore \text{Wt. of NaCl} = 6 - 4 = 2\text{g}$$

9. **Ans (51.00)**

Let initial conc. of HA & NaOH be C_1 &

C_2 mol/L and initial volume of NA = V_1 ml

$$\therefore 5.0 = \text{pK}_a + \log \frac{10C_2}{(C_1V_1 - 10C_2)} \quad \dots(1)$$

$$6.0 = \text{pK}_a + \log \frac{20C_2}{(C_1V_1 - 20C_2)} \quad \dots(2)$$

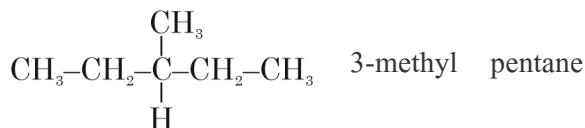
for these we get

$$C_1V_1 = 22.5 C_2$$

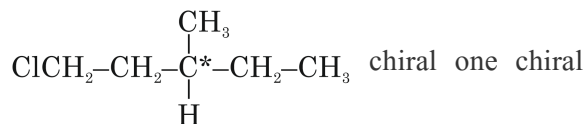
$$\begin{aligned} \therefore 5.0 &= \text{pK}_a + \log \frac{10C_2}{22.5C_2 - 10C_2} \\ &= \text{pK}_a + \log (0.8) \end{aligned}$$

$$\therefore \text{pK}_a = 5.1$$

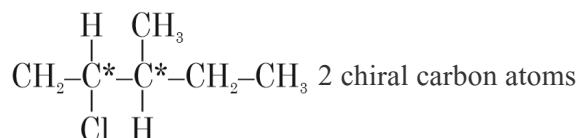
10. Ans (9.00)



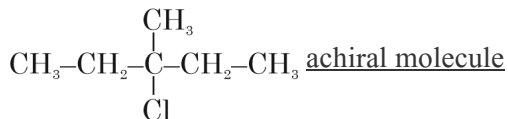
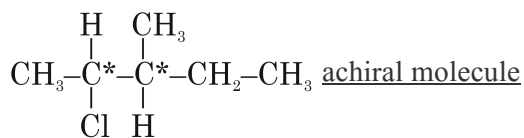
monochlorination, different products are



carbon atom pair of enantiomers



4 isomers. 2 pairs of enantiomers 4 pairs of diastereomers.



X = number of pairs of enantiomers = 3

Y = Number of pairs of diastereomers = 4

Z = number of isomers which are achiral = 2

Total $x + y + z = 9$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (B)

$$\left| x - \frac{\pi}{2} \right| \leq \pi$$

$$\Rightarrow -\pi \leq x - \frac{\pi}{2} \leq \pi$$

$$\Rightarrow -\frac{\pi}{2} \leq x \leq \frac{3\pi}{2}$$

$$\tan^2 x = 3$$

$$\& \tan x > 0$$

$$\Rightarrow \tan x = \sqrt{3}$$

2. Ans (D)

$$f'(x) = 6x^2 - 6(p+2)x + 12p$$

$$D > 0$$

gives two distinct not of $f'(x) = 0$, which gives maxima & minima of $f(x) = 0$

$$36(p+2)^2 - 6 \times 4 \times 12p > 0$$

$$36p^2 + 144p + 144 - 288p > 0$$

$$36p^2 - 144p + 144 > 0$$

$$(p-2)^2 > 0$$

$$p \in \mathbb{R} - \{2\}$$

$$\text{for domain of } f(x), 16 - p^2 > 0 \Rightarrow p \in (-4, 4)$$

$$p \in (-4, 4) - \{2\}$$

3. Ans (A)

$$f'(x) = 3 \left(\frac{\alpha^2 - 64}{\alpha^2 + 2} \right) x^2 - 3$$

f is strictly decreasing on R

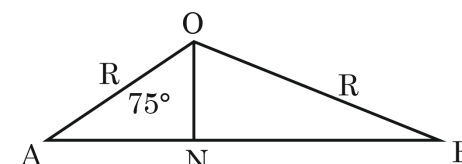
hence

$$\frac{\alpha^2 - 64}{\alpha^2 + 2} \leq 0 \rightarrow \alpha \in [-8, 8]$$

4. Ans (A)

$$AB = 2R \sin \frac{\pi}{12}$$

$$\text{Similarly, } AF = 2R \sin \frac{5\pi}{12}$$



$$\therefore \frac{AB}{AF} + \frac{AF}{AB} = \frac{\sin \frac{\pi}{12}}{\sin \frac{5\pi}{12}} + \frac{\sin \frac{5\pi}{12}}{\sin \frac{\pi}{12}}$$

$$= \tan \frac{\pi}{12} + \cot \frac{\pi}{12}$$

$$= (2 - \sqrt{3}) + (2 + \sqrt{3}) = 4$$

5. Ans (B)

$V = x^3$ and the percent error in measuring x is

$$\frac{dx}{x} \times 100 = \alpha$$

The percent error in measuring volume

$$= \frac{dV}{V} \times 100$$

$$\text{Now, } \frac{dV}{dx} = 3x^2$$

$$\text{or } dV = 3x^2 dx \text{ or } \frac{dV}{V} = \frac{3x^2 dx}{x^3} = 3 \frac{dx}{x}$$

$$\therefore \frac{dV}{V} \times 100 = 3 \frac{dx}{x} \times 100 = 3\alpha$$

6. **Ans (D)**

$$2a^2 + 4b^2 + c^2 = 4ab + 2ca$$

$$\Rightarrow (a - 2b)^2 + (a - c)^2 = 0$$

$$\Rightarrow a = 2b \text{ and } a = c$$

$$\Rightarrow a = 2b = c \text{ or } \frac{a}{1} = \frac{b}{1/2} = \frac{c}{1}$$

$$\Rightarrow \cos B = \frac{7}{8}$$

7. **Ans (A)**

$$y = \frac{1}{x} \Rightarrow \frac{dy}{dx} = -\frac{1}{x^2} \Rightarrow x^2 dy + dx = 0$$

$$\Rightarrow \frac{x^2}{\sqrt{1+x^4}} dy + \frac{dx}{\sqrt{1+x^4}} = 0$$

$$\Rightarrow \frac{dy}{\sqrt{\frac{1}{x^4} + 1}} + \frac{dx}{\sqrt{1+x^4}} = 0$$

$$\Rightarrow \frac{dy}{\sqrt{1+y^4}} + \frac{dx}{\sqrt{1+x^4}} = 0$$

8. **Ans (D)**

$$\sin^2 x \geq \sin^{15} x$$

$$\cos^2 x \geq \cos^{23} x$$

$$\Rightarrow \sin^2 x + \cos^2 x \geq \sin^{15} x + \cos^{23} x$$

$$\Rightarrow \sin^{15} x + \cos^{23} x \leq 1$$

$$\text{LHS : } \sqrt{2 - \cos^{19} x} \geq 1$$

$$\text{so LHS} = \text{RHS} = 1$$

$$\sin^{15} x + \cos^{23} x = 1 \Rightarrow \sin x = 0$$

$$\Rightarrow \cos x = 1 \Rightarrow x = 0, 2\pi, -2\pi$$

9. **Ans (C)**

$$\lim_{x \rightarrow 0} \frac{e \ln(\sec(x) \sec(e^2 x) \dots \sec(e^{40} x))}{e^3 x^2}$$

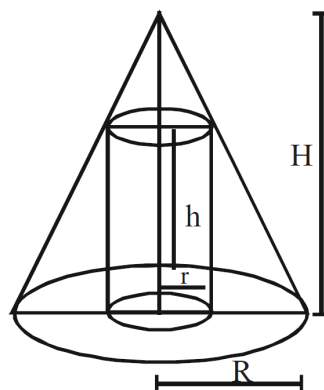
$$\lim_{x \rightarrow 0} \frac{\ln(1 + (\sec(x) \sec(e^2 x) \dots \sec(e^{40} x) - 1))}{e^2 x^2}$$

applying L-H rule

$$L = \frac{e^2 + e^4 + \dots + e^{80}}{2e^2} = \frac{e^2((e^2)^{40} - 1)}{e^2 2(e^2 - 1)} =$$

$$\frac{e^{80} - 1}{2(e^2 - 1)}$$

10. **Ans (B)**



$$\therefore \frac{H}{R} = \frac{H-h}{r} \Rightarrow h = \frac{H}{R}(R-r) \quad \dots(1)$$

$$\therefore V_2 = \pi r^2 h = \pi r^2 \frac{H}{R}(R-r)$$

$$\Rightarrow V_2 = \frac{\pi H}{R}(Rr^2 - r^3)$$

$$\text{put } \frac{dV_2}{dr} = 0 \Rightarrow r = \frac{2R}{3}, h = \frac{H}{3}$$

$$\therefore \frac{V_1}{V_2} = \frac{\frac{1}{3}\pi R^2 H}{\pi \cdot \frac{4R^2}{9} \cdot \frac{H}{3}} = \frac{9}{4}$$

11. **Ans (B)**

Rolle's theorem is applicable on $f(x)$ in $[3, 4]$

$$\Rightarrow f(3) = f(4)$$

$$\Rightarrow 1$$

$$\ln\left(\frac{9+\alpha}{21}\right) = \ln\left(\frac{16+\alpha}{28}\right)$$

$$\Rightarrow (9+\alpha)/21 = (16+\alpha)/28$$

$$\Rightarrow 36 + 4\alpha = 48 + 3\alpha \Rightarrow \alpha = 12$$

$$\text{Now, } f(x) = \ln((x^2 + 12)/7x)$$

$$\Rightarrow f'(x) = (x^2 - 12)/(x(x^2 + 12))$$

$$f'(c) = 0 \Rightarrow c = 2\sqrt{3}$$

$$f''(x) = (-x^4 + 48x^2 + 144)/x^2(x^2 + 12)^2$$

$$\therefore f''(c) = \frac{1}{12}$$

12. **Ans (A)**

$$\frac{b \cos x}{2 \cos 2x - 1} = \frac{b + \sin x}{(\cos^2 x - 3 \sin^2 x) \tan x}$$

$$(1) 2 \cos 2x - 1 \neq 0 \Rightarrow x \neq n\pi \pm \frac{\pi}{6}$$

$$(2) \tan x \neq 0 \Rightarrow x \neq n\pi$$

$$(3) \cos^2 x - 3 \sin^2 x \neq 0 \Rightarrow x \neq n\pi \pm \frac{\pi}{6}$$

$$\text{Also } 2 \cos 2x - 1$$

$$= 2(\cos^2 x - \sin^2 x) - (\cos^2 x + \sin^2 x)$$

$$= \cos^2 x - 3 \sin^2 x$$

Now, the given equation reduces to

$$b \sin x = b + \sin x$$

$$\Rightarrow \sin x = \frac{b}{b-1}$$

$$\text{Now, } -1 \leq \sin x \leq 1$$

$$\Rightarrow -1 \leq \frac{b}{b-1} \leq 1$$

$$\Rightarrow \frac{2b-1}{b-1} \geq 0 \text{ and } \frac{1}{b-1} \leq 0$$

$$\Rightarrow b \in \left(-\infty, \frac{1}{2}\right) \cup (1, \infty) \text{ and } b \in (-\infty, 1)$$

13. **Ans (C)**

$$f'(x) = \cos x - \cos 3x + 2k(2 \sin^2 x)$$

$$f'(x) = 4 \sin^2 x \cos x + 4k \sin^2 x$$

$$f'(x) \geq 0$$

$$\Rightarrow \cos x + k \geq 0$$

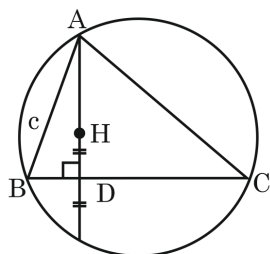
$$\text{i.e. } k \geq 1$$

14. **Ans (B)**

$$AD = c \sin B = 2R \sin B \sin C$$

$$DP = 2R \cos B \cos C$$

$$\therefore AP = 2R(\sin B \sin C + \cos B \cos C) = 4$$



$$2.2\sqrt{2} \cos(B-C) = 4$$

$$\cos(B-C) = \frac{1}{\sqrt{2}}$$

$$\therefore \sec^2(B-C) = 2.$$

15. **Ans (C)**

$$f(x) = x^4 + x^3 + x^2 + x - 1$$

$$f'(x) = 4x^3 + 3x^2 + 2x + 1$$

$$f''(x) = 12x^2 + 6x + 2 > 0 \quad \forall x \in \mathbb{R}$$

$\Rightarrow f'(x)$ is increasing function.

So, $f'(x) = 0$ has exactly one real root.

From Rolle's theorem, $f(x) = 0$ has at most two distinct real roots.

$$\text{Now, } f(x) = -1 < 0.$$

Hence, $f(x) = 0$ only two distinct real roots.

16. **Ans (B)**

$$\tan 2\theta(1 + \cos 2\theta) > 0 \text{ (since } 1 + \cos 2\theta > 0)$$

$$2\theta \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right) \cup \left(2\pi, \frac{5\pi}{2}\right) \cup \left(3\pi, \frac{7\pi}{2}\right)$$

$$\theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$$

17. **Ans (C)**

$$f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$$

$$\Rightarrow f'(x) = 3x^2 + 2x f'(1) + f''(2) \quad \dots(1)$$

$$\Rightarrow f''(x) = 6x + 2f'(1) \quad \dots(2)$$

$$\Rightarrow f'''(x) = 6 \quad \dots(3)$$

put $x = 1$ in equation (1) :

$$f'(1) = 3 + 2f'(1) + f''(2) \quad \dots(4)$$

put $x = 2$ in equation (2) :

$$f''(2) = 12 + 2f'(1) \quad \dots(5)$$

from equation (4) & (5) :

$$-3 - f'(1) = 12 + 2f'(1)$$

$$\Rightarrow 3f'(1) = -15$$

$$\Rightarrow f'(1) = -5 \Rightarrow f''(2) = 2 \quad \dots(2)$$

put $x = 3$ in equation (3) :

$$f'''(3) = 6$$

$$\therefore f(x) = x^3 - 5x^2 + 2x + 6$$

$$f'(x) = 3x^2 - 10x + 2$$

$$f'(2) = -6$$

18. **Ans (C)**

$$\begin{aligned} & \frac{s^2}{\Delta^2} + \frac{(s-a)^2}{\Delta^2} + \frac{(s-b)^2}{\Delta^2} + \frac{(s-c)^2}{\Delta^2} \\ &= \frac{4s^2 - 2s(a+b+c) + a^2 + b^2 + c^2}{\Delta^2} \\ &= \frac{16(a^2 + b^2 + c^2)R^2}{a^2 b^2 c^2} \end{aligned}$$

19. **Ans (A)**

$$T_n = \tan^{-1} \frac{1}{x^2 + (2n-1)x + n^2 - n + 1}$$

$$= \tan^{-1} \frac{1}{1 + (x+n-1)(x+n)}$$

$$= \tan^{-1}(x+n) - \tan^{-1}(x+n-1)$$

$$S_n = \sum_{n=1}^n T_n = \tan^{-1}(x+n) - \tan^{-1}x$$

20. **Ans (B)**

$$p'(1) = 0 \text{ and } p'(2) = 0$$

$$p'(x) = a(x-1)(x-2)$$

$$p(x) = a[(x^3/3) - (3x^2/2) + 2x] + b$$

$$p(1) = 8$$

$$a[(1/3) - (3/2) + 2] + b = 8 \text{ ..(i)}$$

$$p(2) = 4$$

$$a[(8/3) - (3 \times 4/2) + 2 \times 2] + b = 4 \text{ ..(ii)}$$

From equation (i) and (ii)

$$a = 24 \text{ and } b = -12$$

$$p(x) = 24 [(x^3/3) - (3x^2/2) + 2x] - 12$$

hence

$$p(-1) = -104$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (63.50)**

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$= 13 - 6 = 7$$

Now, length of internal angle bisector through vertex

$$C = \frac{2ab}{a+b} \cos \frac{C}{2} = \frac{12}{5} \times \frac{\sqrt{3}}{2} = \frac{6\sqrt{3}}{5} \therefore (A)$$

& length of median through vertex C is

$$\frac{1}{2} \sqrt{2a^2 + 2b^2 - c^2} = \frac{1}{2} \sqrt{26 - 7} = \frac{\sqrt{19}}{2} \therefore (C)$$

2. **Ans (6.00)**

$$2yy' = (2-x)^2 - 2x(2-x)$$

$$(y')_{(1,1)} = -\frac{1}{2}$$

$\therefore$ tangent at P is $y - 1 = -\frac{1}{2}(x - 1)$

$$\Rightarrow 2y + x = 3$$

Solving it with equation of curve

we get $\alpha = \frac{9}{4}, \beta = \frac{3}{8}$

3. **Ans (20.00)**

Let thickness of ice be x cm.

Therefore, net radius of sphere = $(10 + x)$ cm

Volume of sphere $V = (4\pi/3)(10 + x)^3$

$$\Rightarrow dV/dt = 4\pi(10 + x)^2 dx/dt$$

At $x = 5$, $dV/dt = 50 \text{ cm}^3/\text{min}$

$$\Rightarrow 50 = 4\pi \times 225 \times dx/dt$$

$$dx/dt = 1/18\pi \text{ cm/min}$$

$$k = 1/18$$

4. **Ans (4.00)**

$\therefore$ Differentiation of $\sin x$ and $\cos x$ repeat after fourth order

$$\therefore \frac{d^{16}y}{dx^{16}} = \sin x - \cos x$$

$$\Rightarrow f(x) = \frac{d^{17}y}{dx^{17}} = \cos x + \sin x$$

$$\Rightarrow f\left(\frac{\pi}{4}\right) = \cos \frac{\pi}{4} + \sin \frac{\pi}{4}$$

$$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$$

5. **Ans (-0.50)**

$$f(x) = \sin(\ln x) - \cos(\ln x), x > 0$$

$$= \sqrt{2} \sin\left(\ln x - \frac{\pi}{4}\right)$$

$$f'(x) = \frac{\sqrt{2}}{x} \cos\left(\ln x - \frac{\pi}{4}\right)$$

$$= \frac{\sqrt{2}}{x} \sin\left(\frac{\pi}{4} + \ln x\right)$$

$$f'(x) \geq 0 \Rightarrow \sin\left(\frac{\pi}{4} + \ln x\right) \geq 0$$

$$\Rightarrow 0 \leq \frac{\pi}{4} + \ln x \leq \pi \Rightarrow -\frac{\pi}{4} \leq \ln x \leq \frac{3\pi}{4}$$

$$\Rightarrow b - a = \pi$$

$$\Rightarrow \cos(b - a) = -1$$

6. **Ans (1.00)**

$$-\cos(ax + b) = \frac{(x-1)^2}{3} + 1$$

$$\Rightarrow x = 1 \text{ \& } ax + b = \text{any odd multiple of } \pi$$

$$\therefore a + b = \pi$$

7. **Ans (20.00)**

$$f(-7) = -3 \text{ and } f'(x) \leq 2$$

Applying LMVT in $[-7, 0]$, we get

$$\frac{f(-7) - f(0)}{-7 - 0} = f'(c) \leq 2$$

$$\frac{-3 - f(0)}{-7} \leq 2$$

$$f(0) + 3 \leq 14$$

$$f(0) \leq 11$$

Applying LMVT in $[-7, -1]$, we get

$$\frac{f(-7) - f(-1)}{-7 - (-1)} \leq 2$$

$$\Rightarrow f(-1) \leq 9$$

$$\text{Therefore } f(-1) + f(0) \leq 20$$

8. **Ans (3.00)**

$$\Rightarrow 16^{\sin^2 \theta} + 16^{1 - \cos^2 \theta} = 10$$

$$\Rightarrow x + \frac{16}{x} = 10, \text{ where } x = 16^{\sin^2 \theta}$$

$$\Rightarrow x = 2, 8 \Rightarrow 16^{\sin^2 \theta} = 2, 8$$

$$\Rightarrow 2^{4\sin^2 \theta} = 2, 2^3$$

$$\Rightarrow 4\sin^2 \theta = 1, 3$$

$$\Rightarrow \sin^2 \theta = \left(\frac{1}{2}\right)^2, \left(\frac{\sqrt{3}}{2}\right)^2 \Rightarrow \theta = \frac{\pi}{6}, \frac{\pi}{3}$$

$$\therefore \alpha = \frac{\pi}{6}$$

$$10 \cot \frac{\pi}{6} = d$$

$$\Rightarrow d = 10\sqrt{3}$$

9. **Ans (5050.00)**

$$f(x) = (x-1)^{1 \times 100} (x-2)^{2.99} (x-3)^{3.98} \dots$$

$$(x-100)^{100 \times 1}$$

$$\ln f(x) = 100 \ln (x-1) + 2.99 \ln (x-2) + 3.98 \ln$$

$$(x-3) + \dots + 100 \ln (x-100)$$

$$\frac{f'(x)}{f(x)} = \frac{100}{x-1} + \frac{2.99}{x-2} + \frac{3.98}{x-3} + \dots + \frac{100}{x-100}$$

$$\frac{f'(101)}{f(101)} = \frac{100}{100} + \frac{2.99}{99} + \frac{3.98}{98} + \dots + \frac{100}{1}$$

$$= 1 + 2 + 3 + \dots + 100 = 5050$$

10. **Ans (2.00)**

$$4\sin^2 x - 8\sin x + 3 \leq 0$$

$$(2\sin x - 3)(2\sin x - 1) \leq 0$$

$$\Rightarrow \sin x \geq \frac{1}{2}$$

$$\text{hence } x \in \left[\frac{\pi}{6}, \frac{5\pi}{6} \right]$$